Difference-in-Differences: Fundamentals

BUSS975: Causal Inference in Financial Research

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Where we left off

- Lecture 6a, two firms and two years: A treated in year 2, B never.

$$\hat{\delta} = (15 - 10) - (7 - 4) = 2$$

  That was two-way fixed effects. It is also **difference-in-differences**.

- B's change ($+3$) stands in for what A would have done **without treatment**.
- DiD is right exactly when that stand-in is right. This lecture is about when to believe it.

The plan

1. Read Remix ch. 9 on Semmelweis, Snow and Ashenfelter: it is a good story.
2. **Four averages and three subtractions.** The 2×2, and three regressions that compute it exactly.
3. **What makes it causal.** Potential outcomes, parallel trends, no anticipation, and what goes wrong without them.
4. **Card and Krueger.** The study that made it mainstream, and a picture of what regression imputes.
5. **Event studies and falsifications.** Evidence for an assumption you cannot test.
6. **Who gets treated, and why.** Which assignment mechanisms break parallel trends — and which, surprisingly, do not.

- One running question: **what is the counterfactual trend, and why do I believe the control group has it?**

What we are after

A group k is treated at some date; a group U is not. We see Y for both, before and after. Each unit has potential outcomes Y^1 and Y^0; we observe $Y = D\,Y^1 + (1 - D)\,Y^0$.

	Pre	Post
Control U	Y^0 observed	Y^0 observed
Treated k	Y^0 observed	Y^1 observed; Y^0 missing

- **The target** is the average effect on the treated, after treatment: $\text{ATT} = \mathbb{E}[Y_k^1 \mid \text{Post}] - \mathbb{E}[Y_k^0 \mid \text{Post}]$. The second is missing.
- The treated group's own *pre* value? The control group's *post* value?
- *DiD*: the treated group's pre value plus the control group's change. The ATT is then the treated change minus the control change.

2. Four Averages and Three Subtractions

The 2×2 in one line

Let k be the group treated at one date, U a group never treated in the window:

$$\hat{\delta}_{DiD} \;=\; (\bar{Y}_k^{\,post} - \bar{Y}_k^{\,pre}) \;-\; (\bar{Y}_U^{\,post} - \bar{Y}_U^{\,pre})$$

- **Four averages**: each group in each period. **Three subtractions**: two within groups, one across.
- The literature now calls this **the** 2×2 (Goodman-Bacon 2021; Baker et al. 2025). Everything more complex is built from it.
- The within-group subtractions delete anything **fixed about the group**. The across-group subtraction deletes anything **common to the period**. What survives is the effect — and any difference in trends.

The same number from a regression

$$Y_{it} \;=\; \alpha_0 + \alpha_1\,\text{Treat}_i + \alpha_2\,\text{Post}_t + \delta\,(\text{Treat}_i \times \text{Post}_t) + \varepsilon_{it}$$

Four dummy cells, four parameters: OLS fits each cell mean exactly.

	Pre	Post
Control (Treat $= 0$)	$\hat{\alpha}_0$	$\hat{\alpha}_0 + \hat{\alpha}_2$
Treated (Treat $= 1$)	$\hat{\alpha}_0 + \hat{\alpha}_1$	$\hat{\alpha}_0 + \hat{\alpha}_1 + \hat{\alpha}_2 + \hat{\delta}$

Plug the cells into the 2×2:

$$\hat{\delta}_{DiD} = \left[(\hat{\alpha}_0 + \hat{\alpha}_1 + \hat{\alpha}_2 + \hat{\delta}) - (\hat{\alpha}_0 + \hat{\alpha}_1)\right] - \left[(\hat{\alpha}_0 + \hat{\alpha}_2) - \hat{\alpha}_0\right] = (\hat{\alpha}_2 + \hat{\delta}) - \hat{\alpha}_2 = \hat{\delta}$$

Castle doctrine: the data

Cheng and Hoekstra (2013): a balanced panel of 50 states, 2000–2010. Over the decade, 21 states passed **"stand your ground"** laws extending the right to use lethal force in self-defence beyond the home. Outcome: **log homicides** per 100,000.

- To keep it 2×2: keep the 13 states that adopted in **2006** and the 29 that never adopted; drop the other cohorts. Pre = 2000–05, post = 2006–10.

Mean log homicide	Pre	Post	Change
Never adopted	1.252	1.196	−0.055
Adopted in 2006	1.786	1.799	0.013
Difference-in-differences			**0.068**

- Adopters had **higher** homicide rates all along: the level difference is not the effect. Homicide fell in non-adopting states and rose in adopting ones; the gap between those changes is the 2×2: about 7 percent more homicide.

Four ways to the same number

```
k <- subset(castle, is.na(effyear) | effyear == 2006)          # 2006 cohort + never treated
k$treat <- !is.na(k$effyear); k$post <- k$year >= 2006
m <- with(k, tapply(l_homicide, list(treat, post), mean))       # 1. the 2x2
(m[2, 2] - m[2, 1]) - (m[1, 2] - m[1, 1])
feols(l_homicide ~ treat * post, k, cluster = ~sid)            # 2. interaction OLS
feols(l_homicide ~ treat:post | sid + year, k, cluster = ~sid) # 3. two-way fixed effects
w <- ...  # each state's post-period mean minus its pre-period mean
feols(dbar ~ treat, w, cluster = ~sid)                          # 4. the long difference
```

	2×2	Interaction OLS	TWFE	Long difference
$\hat{\delta}$	0.0682359	0.0682359	0.0682359	0.0682359

- **Three regressions, one arithmetic.** With two groups and two periods, unit and year fixed effects are just Treat_i and Post_t in disguise.
- This is why "difference-in-differences" and "two-way fixed effects" were synonyms for thirty years. They stop being synonyms when treatment dates differ across units — next lecture.

Why regression, then?

- Not for the point estimate: that is fixed by the arithmetic.
- For **standard errors** — and for adding covariates, more periods, more groups. All of which, we will see, come with strings attached.

3. What Makes It Causal

From subtraction to causality: potential outcomes

Write the 2×2 with expectations and swap in potential outcomes using the switching equation $Y = D\,Y^1 + (1 - D)\,Y^0$:

$$\hat{\delta}_{DiD} = (\mathbb{E}[Y_k|\text{Post}] - \mathbb{E}[Y_k|\text{Pre}]) - (\mathbb{E}[Y_U|\text{Post}] - \mathbb{E}[Y_U|\text{Pre}])$$

$$= (\mathbb{E}[Y_k^1|\text{Post}] - \mathbb{E}[Y_k^0|\text{Pre}]) - (\mathbb{E}[Y_U^0|\text{Post}] - \mathbb{E}[Y_U^0|\text{Pre}])$$

- Only one of the four cells is treated: **group k, after.** Every other cell is Y^0.
- Nothing here is a causal effect yet. A causal effect is $Y^1 - Y^0$ **for the same group at the same time**. Each pair above compares either two groups, or two dates.

Adding a zero

$$
\begin{aligned}
\hat{\delta}_{DiD} &= \underbrace{(\mathbb{E}[Y_k^1|\text{Post}] - \mathbb{E}[Y_k^0|\text{Pre}]) - (\mathbb{E}[Y_U^0|\text{Post}] - \mathbb{E}[Y_U^0|\text{Pre}])}_{\text{Switching equation}} \\
&\quad + \underbrace{\mathbb{E}[\mathbf{Y_k^0}|\text{Post}] - \mathbb{E}[\mathbf{Y_k^0}|\text{Post}]}_{\text{Adding zero}} \\
&= \underbrace{\mathbb{E}[Y_k^1|\text{Post}] - \mathbb{E}[Y_k^0|\text{Post}]}_{\textbf{ATT}} \\
&\quad + \underbrace{(\mathbb{E}[Y_k^0|\text{Post}] - \mathbb{E}[Y_k^0|\text{Pre}]) - (\mathbb{E}[Y_U^0|\text{Post}] - \mathbb{E}[Y_U^0|\text{Pre}])}_{\textbf{non-parallel trends bias}}
\end{aligned}
$$

- The first term is the average effect *on the treated*, *in the post period*.
- The second term is a selection bias — in *changes* of Y^0.

What parallel trends says, and does not

- Parallel trends is the assumption that the second term is zero:

$$\mathbb{E}[Y_k^0|\text{Post}] - \mathbb{E}[Y_k^0|\text{Pre}] \;=\; \mathbb{E}[Y_U^0|\text{Post}] - \mathbb{E}[Y_U^0|\text{Pre}]$$

- It is a statement about a **counterfactual**: how the treated group's Y^0 *would have* moved. **It cannot be tested.**
- It says nothing about **levels**. Adopters can have twice the homicide rate. Fixed differences are subtracted out. What must match is the *change*.
- It is **specific to the comparison and the baseline you chose.** Change the control group, or the pre period, and you have made a different assumption.
- It is a statement in a particular **scale**: parallel in levels is not parallel in logs. Pick the scale by the economics, before you see the estimate.

- Three assumptions, in total: **parallel trends**, **no anticipation** (next), and SUTVA — one version of the treatment, no spillovers onto the controls.

A numerical example: parallel trends holds

Two groups, 1980–1990, group 1 treated from 1986. The simulation knows both potential outcomes; the data show only one.

	Observed Y		Untreated Y^0	
Average	Group 1	Group 2	Group 1	Group 2
Pre (1980–85)	6.06	6.07	6.06	6.07
Post (1986–90)	41.56	11.59	11.57	11.59
Change	35.51	5.52	5.51	5.52

- **DiD** $= 35.51 - 5.52 = 29.99$, from the observed columns.
- **ATT** $= 41.56 - 11.57 = 30.0$: group 1's observed post outcome minus its *untreated* post outcome.
- **Non-parallel trends bias** $= 5.51 - 5.52 = -0.01$, the 2×2 on Y^0. And $29.99 = 30.0 + (-0.01)$.

Same data, different counterfactual

Change only group 1's post-period Y^0.

	Observed Y		Untreated Y^0	
Average	Group 1	Group 2	Group 1	Group 2
Pre (1980–85)	6.06	6.07	6.06	6.07
Post (1986–90)	41.56	11.59	**36.60**	11.59
Change	35.51	5.52	**30.55**	5.52

- **DiD** $= 29.99$. Unchanged — it uses no counterfactual.
- **ATT** $= 41.56 - 36.60 = 4.96$. **Bias** $= 30.55 - 5.52 = 25.03$. Group 1 was going to rise anyway.
- **The point.** Two worlds, identical data, same $\hat{\delta}$, ATT of 30 in one and 5 in the other. The data cannot tell you which world you are in. Only an argument about *why* group 1 was treated can — Section 6.

No anticipation: what it literally says

The treated group is untreated until it is treated: $Y = Y^0$ for k in the pre period. Suppose it fails, so $Y_k = Y_k^1$ in *both* periods. Add $\pm\mathbb{E}[Y_k^0|\text{Post}]$ and $\pm\mathbb{E}[Y_k^0|\text{Pre}]$:

$$\begin{aligned}\hat{\delta}_{DiD} &= \big(\mathbb{E}[Y_k^1|\text{Post}] - \mathbb{E}[\mathbf{Y_k^1}|\text{Pre}]\big) - \big(\mathbb{E}[Y_U^0|\text{Post}] - \mathbb{E}[Y_U^0|\text{Pre}]\big) \\ &= \underbrace{\mathbb{E}[Y_k^1|\text{Post}] - \mathbb{E}[Y_k^0|\text{Post}]}_{\text{ATT}_k(\text{Post})} \\ &\quad + \underbrace{\big(\mathbb{E}[Y_k^0|\text{Post}] - \mathbb{E}[Y_k^0|\text{Pre}]\big) - \big(\mathbb{E}[Y_U^0|\text{Post}] - \mathbb{E}[Y_U^0|\text{Pre}]\big)}_{\text{non-parallel trends bias}} \\ &\quad - \underbrace{\big(\mathbb{E}[Y_k^1|\text{Pre}] - \mathbb{E}[Y_k^0|\text{Pre}]\big)}_{\text{ATT}_k(\text{Pre})} = \text{ATT}_k(\text{Post}) + \text{PT bias} - \text{ATT}_k(\text{Pre})\end{aligned}$$

- DiD recovers the ATT only if parallel trends holds **and** $\text{ATT}_k(\text{Pre}) = 0$.
- With **constant** effects DiD is **zero**; with **growing** effects it is the *growth* in the effect, not the effect.

No anticipation, simulated

1,000 firms, 1990–93, half treated from 1991 onward. Y^0 has a firm effect and a common trend. Two effect paths: **constant** $(10, 10, 10)$ and **dynamic** $(10, 20, 30)$.

	Baseline = 1990 (clean)		Baseline = 1991 (already treated)	
	constant	dynamic	constant	dynamic
$\hat{\delta}_{DiD}$	10.03	20.03	0.06	15.06
ATT over the post period	10	20	10	25

- Clean baseline: DiD hits the ATT in both columns (10 and 20).
- Treated baseline, constant effect: **zero**. The effect is in both periods and cancels.
- Treated baseline, dynamic effect: $25 - 10 = 15$. You estimate the *change* in the effect and call it the effect.

Dating the treatment

- **The baseline must be clean.** If a group was treated even briefly in a period, that period belongs to *post*. Err on the side of starting treatment early.
- **Announcement vs. enforcement.** If units plausibly respond when a law is *passed* rather than when it takes effect, make the baseline the period before passage.

- Be aware of what that choice changes:
  - the **ATT** — it now averages over an "announced but not enforced" period as well;
  - the **parallel trends assumption** — it is always relative to a baseline, and you just moved it.
- None of this is right or wrong in the abstract. It is a design decision; make it on purpose, and say what you did.

Avoid already-treated controls

Suppose control U was treated *before* the window: $Y_U = Y_U^1$ in both periods:

$$\begin{aligned}
\hat{\delta}_{DiD} &= \big(\mathbb{E}[Y_k^1|\text{Post}] - \mathbb{E}[Y_k^0|\text{Pre}]\big) - \big(\mathbb{E}[\mathbf{Y_U^1}|\text{Post}] - \mathbb{E}[\mathbf{Y_U^1}|\text{Pre}]\big) \\
&= \underbrace{\mathbb{E}[Y_k^1|\text{Post}] - \mathbb{E}[Y_k^0|\text{Post}]}_{\text{ATT}_k} \\
&\quad + \underbrace{\big(\mathbb{E}[Y_k^0|\text{Post}] - \mathbb{E}[Y_k^0|\text{Pre}]\big) - \big(\mathbb{E}[Y_U^0|\text{Post}] - \mathbb{E}[Y_U^0|\text{Pre}]\big)}_{\text{non-parallel trends bias}} \\
&\quad - \underbrace{\big[\big(\mathbb{E}[Y_U^1|\text{Post}] - \mathbb{E}[Y_U^0|\text{Post}]\big) - \big(\mathbb{E}[Y_U^1|\text{Pre}] - \mathbb{E}[Y_U^0|\text{Pre}]\big)\big]}_{\Delta\text{ATT}_U\text{: how the } \textit{control's} \text{ effect changed}}
\end{aligned}$$

- So $\hat{\delta}_{DiD} = \text{ATT}_k + \text{PT bias} - \Delta\text{ATT}_U$. A **constant** control effect differences out; a **fading or growing** one looks like an effect of the opposite sign.
- Nobody would *choose* such a control. With staggered dates, two-way fixed effects does it **silently** — next lecture.

4. Card and Krueger

New Jersey vs. Pennsylvania, 1992

- Late 1991: New Jersey will raise its minimum wage from \$4.25 to \$5.05 in April 1992. Pennsylvania, next door, will not.
- Card and Krueger phoned every fast-food restaurant they could find in both states — **February** (before) and **November** (after) — and asked about wages and employment. They ran their own survey because no dataset existed.

- **First, show the policy had bite.** In November, the NJ wage distribution has a spike at exactly \$5.05: a large mass of workers was pushed up to the new minimum. In PA, no spike. The treatment *happened* to the treated.
- Always do this before looking at the outcome. If the policy did not move the thing it was meant to move, there is no reason for anything downstream to have moved either.

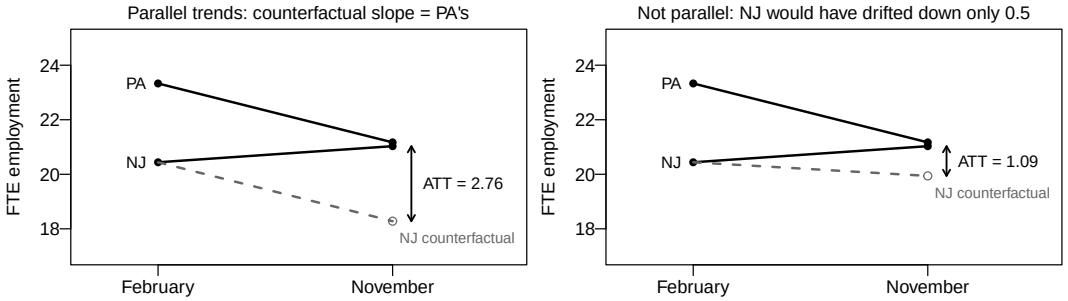
The result: four averages, three subtractions

Full-time-equivalent employment per store (Card and Krueger 1994, Table 3):

	PA	NJ	NJ − PA
FTE, February (before)	23.33 (1.35)	20.44 (0.51)	−2.89 (1.44)
FTE, November (after)	21.17 (0.94)	21.03 (0.52)	−0.14 (1.07)
Change	−2.16 (1.25)	+0.59 (0.54)	+**2.76** (1.36)

- Competitive theory says employment should **fall** when the wage floor rises. The estimate is $+2.76$ FTE per store, $t \approx 2$.
- The reaction split the profession. Whatever one believes about the economics, the paper put a 2×2 in front of everyone, and the method's use in economics took off from here (Currie, Kleven and Zwiers 2020).
- Note how a reader can check every number in this table. That is a feature.

What the regression imputes



- **Left.** The regression draws NJ's missing November point by giving it PA's slope. The gap between NJ's actual November and that imputed point is $\hat{\delta} = 2.76$.
- **Right.** Same four observed points, same $\hat{\delta} = 2.76$. But here NJ's true counterfactual fell by only 0.5, so the true ATT is 1.09. **The estimate does not know which picture it is in.**

5. Event Studies and Falsifications

The event-study regression

The 2×2 interacts treatment with *one* post dummy. Interact it with **every period** except the baseline instead:

$$Y_{it} \;=\; \alpha_i + \lambda_t + \sum_{\tau \neq -1} \delta_\tau \left(D_i \times \mathbf{1}\{t - t^* = \tau\} \right) + \varepsilon_{it}$$

- t^* is the treatment date, τ the distance from it. The baseline $\tau = -1$ is omitted: it must be untreated, which is no anticipation again.
- **Each $\hat{\delta}_\tau$ is its own** 2×2 against the baseline. For $\tau = -3$:

$$\hat{\delta}_{-3} = (\bar{Y}_k^{\tau=-3} - \bar{Y}_k^{\tau=-1}) - (\bar{Y}_U^{\tau=-3} - \bar{Y}_U^{\tau=-1})$$

- Sometimes called "long differences": every period is compared to the same fixed baseline, however far away.

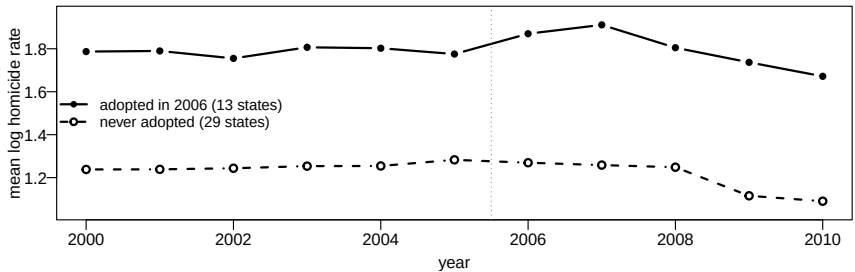
What the coefficients measure

Substitute potential outcomes. Before treatment, no anticipation makes $\text{ATT}_\tau = 0$, so

$$\hat{\delta}_{-3} \;=\; (\mathbb{E}[Y_k^0|\tau=-3] - \mathbb{E}[Y_k^0|\tau=-1]) - (\mathbb{E}[Y_U^0|\tau=-3] - \mathbb{E}[Y_U^0|\tau=-1])$$

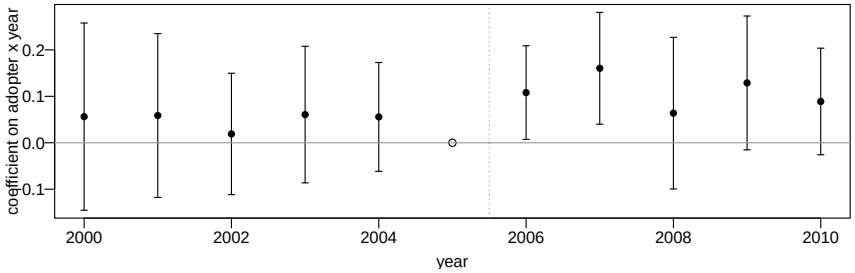
- **All four terms are observable.** A pre-period coefficient is the *differential pre-trend* in untreated outcomes — which is a fact about the data, **not the parallel trends assumption.** Parallel trends is about the post period, where one term is counterfactual.
- $\hat{\delta}_{-3} = +1.5$ means: from $\tau = -3$ to the baseline, the treated group's outcome rose 1.5 more than the control's.
- **After treatment**, $\hat{\delta}_{+2}$ is the ATT at $\tau = +2$ — *if* parallel trends holds from the baseline to $\tau = +2$, there is no anticipation, and the control is never- or not-yet-treated.

Castle doctrine: the raw data first



- Cheng and Hoekstra plotted exactly this, one panel per cohort, so readers could compare the two paths with their own eyes.
- Two flat, roughly parallel lines for six years; then the adopters move up while the others drift down. Note also what the picture does *not* show: how either line would have moved in the other state of the world.

Castle doctrine: the coefficients



- Each point is a 2×2 against 2005, with a 95% interval, standard errors clustered by state. The hollow point is the omitted baseline.
- Pre-period coefficients: 0.02 to 0.06, none distinguishable from zero. Post: 0.11 in 2006, 0.16 in 2007, and positive throughout.
- The 2×2 of 0.068 is a weighted average of the five post coefficients, less a weighted average of the pre ones.

Presenting an event study

- Plot the coefficients with their intervals. Two conventions: **disconnected** points with whiskers, or a **connected** line with a shaded band.
- A case for disconnected: the band seems to pinch to zero at the baseline, which suggests precision where nothing was estimated; and you can count the coefficients.
- Neither is correct. Choose what your audience reads correctly — and make it beautiful enough that they want to look.

- What a flat pre-period buys you is **suggestive** evidence, not a test. Parallel *pre*-trends are not required for identification, and parallel *post*-trends cannot be checked. Section 6 shows a case where the pre-trends break and parallel trends holds, and one where the pre-trends look fine and it does not.

Falsification: the logic

- A theory earns credibility by **excluding alternatives** (Popper 1959). Your DiD result has two explanations: the treatment did it, or something else did.
- A falsification takes the *something else* seriously: if it were true, what else would we see? Then go and look.
- Two designs:
  - **Same outcome, alternative group** — a group the treatment cannot touch, but the alternative story would.
  - **Same group, alternative outcome** — an outcome the treatment cannot touch, but the alternative story would.

- Passing a falsification does not prove parallel trends; failing one does not prove the main result wrong. Both are **evidence**, and they weigh more than another asterisk.

Falsification: two examples

- **Same outcome, alternative group.** You find the minimum wage reduced low-wage employment. Alternative: the whole state economy was sinking. Then **high-wage** workers — untouched by the floor — should have lost jobs too. Re-run the 2×2 on them. If they did, your main result has a rival explanation.
- **Same group, alternative outcome.** Cigarette taxes reduce heart-attack admissions. Alternative: that increased preventative healthcare expanded at the same time, in the same states. Then admissions for **appendicitis and fractures** — nothing to do with smoking — should have fallen as well. If they did not, the rival loses.

- The design needs a rival theory that makes a **prediction the treatment does not**.

Inference

- Most DiD panels have **many periods**, treatment that varies **at the group level** (a state, an industry), and outcomes that are **serially correlated**. Bertrand, Duflo and Mullainathan (2004) showed that conventional standard errors then over-reject badly.
- Three fixes, from their paper:
  1. **Cluster** at the level treatment is assigned — the state, not the restaurant.
  2. **Aggregate** to one pre and one post observation per unit: the long difference. Serial correlation has nowhere to live.
  3. **Block bootstrap**: resample whole states.

- With **few clusters**, or few *treated* clusters, clustering itself over-rejects. Castle doctrine has 13 adopters. Lecture 6a's wild cluster bootstrap applies; with a single treated unit, randomization inference may be an option (Buchmueller, DiNardo and Valletta 2011).

DiD in the courtroom

A prosecutor has three things: a **claim** of guilt, **evidence** for it, and a **motive**. The claim is not evidence, and a case with no motive is hard to believe beyond reasonable doubt.

- Your regression table is the **claim**.
- Five elements of a strong DiD paper:
  1. **Bite**: the policy moved what it was meant to move.
  2. **Event studies**: the two groups moved together before.
  3. **Falsifications**: the rival explanations were given a chance to show up, and did not.
  4. **Main results**.
  5. **Mechanism**: *why* the effect appears — the motive.

Medicaid and mortality (Miller, Johnson and Wherry 2021)

The 2014 Medicaid expansion was adopted by some states and not others. Outcome: annual mortality of **near-elderly** adults (55–64), from death records linked to a large federal survey.

- **Bite** (three event studies): eligibility up, enrollment up, the uninsured share *down* — so enrollment was not just people switching from private cover.
- **Falsification, same outcome, alternative group:** adults **65 and over** are on Medicare and cannot be affected. Their enrollment and mortality: no change.
- **Main result:** mortality falls 0.13 percentage points, 9.3% of the mean, growing over time. The intervals are wide; the post coefficients are each different from zero but not from each other, and a linear trend cannot be rejected.
- **Mechanism:** the decline is in disease-related deaths — conditions that treatment, once you have access to it, prevents.

- Could still be wrong: the counterfactual is unobserved. But the layered evidence leaves a sceptic with few places to stand. Read it for how the argument is built.

6. Who Gets Treated, and Why

The assignment mechanism

- Now the question that decides whether the assumption is plausible: **why** were these units treated and not those?
- Ghanem, Sant'Anna and Wüthrich (2024) and Marx, Tamer and Tang (2024) ask which mechanisms are compatible with parallel trends. **Good news**: not sufficient to violate it. **Bad news**: sufficient.

Good news: not by itself a violation	Bad news: a violation
a **common constant trend** in Y^0 selection on the **baseline** Y^0 selection on **fixed effects** selection on **observables** (conditional PT) selection under **imperfect foresight**	the **Perfect Doctor**: selection on the gain $Y^1 - Y^0$ (Roy 1951; "essential heterogeneity", Heckman et al. 2006)

- This knowledge comes from **outside the data**: institutional detail about how units came to be treated.

Good news: a common constant trend

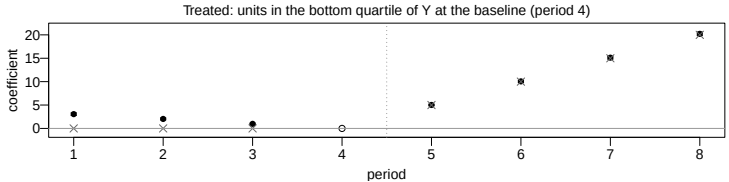
The one case where parallel trends **cannot** fail, however the treated were chosen.

- Parallel trends compares two *changes* in the untreated outcome:

$$\underbrace{\mathbb{E}\big[Y^0_{\text{Post}} - Y^0_{\text{Pre}} \mid \text{treated}\big]}_{\text{how the treated would have moved}} = \underbrace{\mathbb{E}\big[Y^0_{\text{Post}} - Y^0_{\text{Pre}} \mid \text{control}\big]}_{\text{how the control did move}}$$

- Suppose **every** unit's untreated outcome moves by the same amount, Δ, from pre to post. Then both sides equal Δ, whoever is in the treated group.
- Levels can differ as much as you like, and the level can be the reason a unit was treated. Only the *change* has to be the same.
- This is the world assumed by $Y_{it} = \alpha_i + \lambda_t + \delta D_{it} + \varepsilon_{it}$: the year effect λ_t *is* the common move. A strong claim.
- The simulations that follow **drop** it: units carry different trends, and we ask which ways of choosing the treated still leave parallel trends intact.

Good news: selection on the baseline Y^0



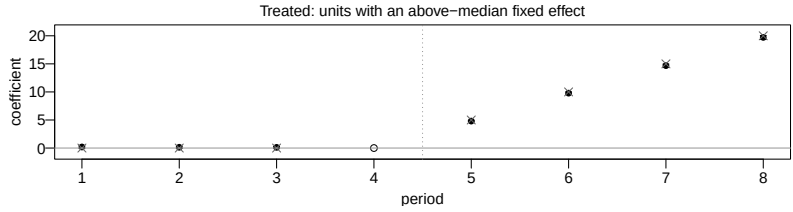
Betty earns below a threshold and takes the Python course; Ronnie does not. Should we worry whether the different levels of the treated and control groups impact trends?

- 2,000 workers, 8 periods. **Treatment effect**: zero before period 5, then $5, 10, 15, 20$ ($\times$ = truth). **Untreated outcome** Y^0: a worker fixed effect α_i (a permanent level), a random walk, and a trend that differs across **four groups** of workers (slopes 0.5 to 2, drawn at random). No common constant trend; but group is unrelated to treatment, so trends are parallel *on average*.
- **Post-coefficients on the truth.** Ghanem, Sant'Anna and Wüthrich (2024): selecting on the baseline Y^0 does not violate parallel trends.

Baseline Y^0: broken pre-trends, not a broken assumption

- **But the pre-coefficients are not zero**, although $\mathbb{E}[Y^0]$ trends are parallel by group. The baseline is both the point of selection and the omitted category: the treated were picked for being low *then*, so every other period looks high by comparison. A **mechanical** artefact.
- **No solution, because no problem.** The post-coefficients are unbiased. The casualty is the pre-trend *test*: it rejects, and you wrongly conclude parallel trends failed.
- The way out is **institutional knowledge of the assignment mechanism**. If the agency targeted the lowest-scoring schools, you know in advance: parallel trends intact, pre-trends disrupted. Falsifications (Section 5) then matter more than pre-trends.
- **Caveat**: if the low baseline is a **transitory dip** that reverts on its own — Ashenfelter's dip — trends do differ. Ask the institution, not the data.

Good news: selection on fixed effects



Attributes that are permanent for a unit are its **fixed effects**, carried wherever and whenever it goes.

- Same data. Now **treatment goes to the workers whose fixed effect is above the median**: only certain *types* enter the programme.
- Does not violate parallel trends, and — unlike selection on Y^0 — does **not** break the pre-trends either. Pre-coefficients at zero, post-coefficients on the truth: permanent traits are differenced out.
- Again: learn, ahead of time, whether the mechanism selected on fixed effects.

Good news: selection on observables

Being selected for the programme because of your GPA or your age is not itself a violation of parallel trends.

- Parallel trends may hold only *within* strata of the covariates: **conditional parallel trends**. Workers of the same age and GPA move in parallel; the treated are simply older, or have higher GPAs, than the controls.
- Then the covariates must be in the model, so that after accounting for them the untreated trends of treated and control are identical. How to do that correctly — it is not as simple as adding X_i to the regression — is next lecture's subject.
- Not a guarantee. Selection on observables and non-parallel trends can coexist; the claim is only that this mechanism is not a problem *in and of itself*.

- The link to Lecture 3: this is the panel version of unconfoundedness. There we assumed the *level* of Y^0 was independent of treatment given X; here only its *change* has to be.

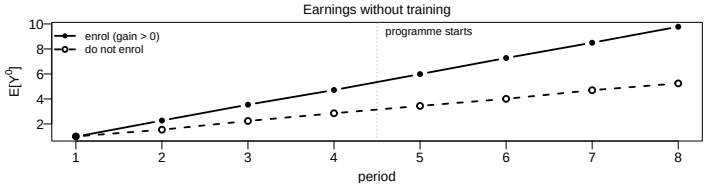
Good news: selection under imperfect foresight

If units are assigned to treatment **without fully anticipating their future outcomes**, selection cannot depend on the expected treatment effect, and parallel trends can hold.

- The leading case: assignment on *past or present* performance, with no knowledge of how the programme will change future earnings. Whatever drives selection is then already in the fixed effect or the baseline — the good-news cases above.
- Most relevant where treatment is not self-selected on anticipated gains: an agency applies a rule, a firm is caught by a threshold, a manager acts on last year's numbers.
- Ghanem, Sant'Anna and Wüthrich (2024): imperfect foresight does not by itself mangle parallel trends. It is also the realistic case more often than economists like to admit.

- Five mechanisms, five ways assignment can interact with Y^0 and leave parallel trends standing. What if none of them applies — if units choose on the gain itself, and get it right?

Bad news: the Perfect Doctor



A training programme. Workers enrol **iff it will help them**, and they are right.

- **Who gains?** Older, high-GPA workers. **Whose earnings rise faster anyway?** The same workers: the market pays more for age and GPA every year.
- So choosing on the gain *is* choosing the steeper path. The picture shows earnings *without* training: enrollees pull ahead regardless.
- DiD credits that head start to the programme ("Adding a zero", Section 3):

DiD	=	ATT	+	non-parallel trends bias
5.27	=	2.80	+	2.47

Is the Perfect Doctor the rule or the exception?

- **Not guaranteed.** Imperfect foresight does not mangle parallel trends, and humans are not good at working out what is best for them in counterfactual states
- **Nevertheless.** Firms issue, list, adopt and enter when they expect to gain: the Perfect Doctor by construction. This is why finance looks for treatments imposed on firms, by a state law, a regulator's threshold, an index rule or a court, rather than chosen by them. The next slide lists them.

Difference-in-differences in finance

- **Where the 2×2 comes from.** State laws (Bertrand and Mullainathan 2003: anti-takeover laws and the "quiet life"); regulatory thresholds and exemptions; exchange and index rules; court rulings; natural disasters. Treatment is assigned to firms by *geography or rule*, not by choice.
- **Bite.** Did the law actually bind? Which firms were exposed? If exposure is a choice, you are back to the Perfect Doctor.
- **Clusters.** Treatment at the state level means clustering at the state level, and often few treated states. Bertrand, Duflo and Mullainathan was written about exactly these designs.
- **Falsification.** Firms incorporated elsewhere but headquartered in the state; outcomes the law cannot reach; a placebo year.

- And the question from this lecture: **why these states, these years?** A law passed because a state's firms were already struggling is selection on the trend.

Conclusion

1. **DiD is four averages and three subtractions.** Three regressions compute it exactly; regression adds standard errors, not identification.
2. **It identifies the ATT in the post period, plus a bias**: the 2×2 on Y^0. Parallel trends is the assumption that this bias is zero. It is about a counterfactual and **cannot be tested.**
3. **No anticipation** means the baseline is untreated. Violate it and you estimate the *change* in the effect. **Already-treated controls** subtract the change in *their* effect.
4. **Event studies** show whether the groups moved together *before*; falsifications give rival explanations a chance to appear. Both are evidence, neither is proof.
5. **The assignment mechanism decides.** Selection on fixed traits or baseline levels can be fine; selection on gains, or on a dip that will revert, is not. Learn *why* units were treated before you estimate anything.

- **Next:** more than one treatment date, covariates, and what two-way fixed effects does when the 2×2's silently multiply.

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